

MURUN, Mongolia

WMO: 442310

Lat: 49.6383N

Lon: 100.1669E

Elev: 1283

StdP: 86.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.8	-32.3	-38.5	0.1	-34.2	-35.9	0.1	-31.7	8.8	-13.1	7.5	-13.4	0.2	20	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.5	15.8	27.4	15.1	25.7	14.5	17.8	25.4	16.7	24.1	15.8	23.0	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	12.7	20.9	14.1	11.7	19.4	13.1	11.0	19.0	55.9	25.5	52.1	24.1	49.2	23.1	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.8	6.7	-36.1	33.1	3.4	2.4	-38.6	34.9	-40.6	36.3	-42.5	37.6	-45.0	39.4		
(5)			-36.2	19.6	3.4	1.7	-38.7	20.9	-40.7	21.9	-42.6	22.8	-45.1	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-21.4	-15.8	-5.4	4.1	10.6	16.7	18.9	16.6	9.9	1.3	-9.7	-17.9	(6)
(7)		DBStd	14.73	5.67	6.10	6.20	4.92	4.70	3.72	3.09	3.24	4.07	4.77	6.52	6.49	(7)
(8)		HDD10.0	4190	973	723	477	185	50	2	0	1	51	271	591	864	(8)
(9)		HDD18.3	6512	1232	957	736	428	242	71	29	71	253	529	841	1123	(9)
(10)		CDD10.0	812	0	0	0	6	69	204	277	207	48	0	0	0	(10)
(11)		CDD18.3	93	0	0	0	0	2	23	48	18	1	0	0	0	(11)
(12)		CDH23.3	1023	0	0	0	2	69	294	448	197	14	0	0	0	(12)
(13)		CDH26.7	259	0	0	0	0	11	71	129	47	1	0	0	0	(13)
(14)	Wind	WSAvg	2.0	1.1	1.4	2.4	3.0	2.4	2.1	1.9	2.0	1.9	1.7	1.4	(14)	
(15)	Precipitation	PrecAvg	227	2	1	1	7	16	45	67	58	19	6	3	2	(15)
(16)		PrecMax	397	18	6	8	34	54	179	167	163	60	27	14	8	(16)
(17)		PrecMin	133	0	0	0	0	2	2	5	2	1	0	0	0	(17)
(18)		PrecStd	67	3	1	2	8	12	31	35	32	14	6	3	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.2	1.3	13.8	21.7	28.2	31.2	32.9	30.7	25.1	17.0	7.5	-1.9	(19)
(20)			MCWB	-6.8	-2.9	5.3	8.4	13.3	15.3	18.7	16.2	13.7	7.8	1.9	-4.9	(20)
(21)		2%	DB	-7.9	-1.5	10.0	17.8	25.3	28.7	30.0	27.6	22.3	14.1	4.4	-4.6	(21)
(22)			MCWB	-9.9	-4.9	2.8	7.3	11.6	14.7	16.8	15.4	12.2	6.2	0.1	-7.2	(22)
(23)		5%	DB	-10.7	-4.0	7.2	15.4	22.6	26.7	27.7	25.3	20.0	11.7	1.8	-6.8	(23)
(24)			MCWB	-12.3	-6.8	0.8	5.9	10.3	14.0	16.1	14.9	10.9	4.8	-1.8	-8.9	(24)
(25)		10%	DB	-13.0	-6.5	4.5	13.3	20.2	24.6	25.7	23.4	17.9	9.3	-0.4	-9.0	(25)
(26)			MCWB	-14.2	-8.8	-0.8	4.9	9.3	13.2	15.6	14.4	9.8	3.4	-3.4	-10.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.5	-2.6	5.6	9.6	14.6	17.7	20.1	18.5	15.4	8.4	2.1	-4.7	(27)
(28)			MCDWB	-4.3	0.9	13.2	19.1	25.7	26.7	29.8	26.0	22.9	15.9	6.7	-2.1	(28)
(29)		2%	WB	-9.9	-4.8	3.0	7.8	12.4	16.2	18.6	16.9	13.2	6.6	0.2	-7.1	(29)
(30)			MCDWB	-7.9	-1.6	9.6	16.9	23.4	25.0	27.1	24.2	20.0	13.5	4.2	-4.7	(30)
(31)		5%	WB	-12.2	-6.8	1.0	6.4	11.1	15.2	17.4	15.9	11.7	5.2	-1.7	-8.9	(31)
(32)			MCDWB	-10.7	-4.1	6.7	14.5	21.2	23.2	24.6	23.1	18.6	11.1	1.6	-6.8	(32)
(33)		10%	WB	-14.2	-8.8	-0.7	5.0	9.7	14.3	16.4	15.1	10.3	3.7	-3.3	-10.8	(33)
(34)			MCDWB	-13.0	-6.6	4.4	12.9	19.1	22.3	23.3	21.7	16.7	9.0	-0.4	-9.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.7	13.4	13.7	13.1	13.4	12.3	11.2	11.6	13.0	12.7	10.4	9.4	(35)
(36)			MCDBR	12.1	15.3	15.9	16.8	17.6	15.8	14.7	15.1	16.5	16.6	12.3	10.4	(36)
(37)			MCWBR	10.5	12.4	10.4	9.6	8.9	6.6	5.6	6.2	8.4	10.0	8.8	8.8	(37)
(38)		5% WB	MCDWB	11.9	15.0	15.4	15.6	16.0	12.9	12.7	12.4	14.8	15.5	11.8	10.4	(38)
(39)			MCWBR	10.4	12.3	10.5	9.6	8.9	6.1	5.6	5.8	7.9	9.7	8.8	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.237	0.262	0.310	0.350	0.371	0.368	0.360	0.341	0.302	0.283	0.261	0.240	(40)	
(41)		taud	2.330	2.290	2.234	2.246	2.224	2.344	2.414	2.441	2.485	2.458	2.355	2.300	(41)	
(42)		Ebn at Noon	838	892	896	893	885	890	892	896	903	862	792	772	(42)	
(43)		Edh at Noon	65	88	112	125	134	120	110	102	87	74	63	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.64	2.74	4.26	5.53	6.41	6.46	6.14	5.32	4.42	2.96	1.74	1.27	(44)	
(45)		RadStd	0.07	0.12	0.15	0.29	0.36	0.34	0.28	0.22	0.22	0.13	0.08	0.07	(45)	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page